

Rajjat Dadwal, PhD

MACHINE LEARNING · AGENTIC AI · LLM FINE-TUNING

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Profile

Senior Research Scientist with a PhD in Artificial Intelligence and 8 years of experience building and fine-tuning machine learning systems, including hands-on daily use of agentic coding assistants (Claude Code) for autonomous, multi-step engineering workflows. Track record designing LLM evaluation frameworks and fine-tuning LLMs for applied tasks, including distributed training with DeepSpeed. Combines a research-grade understanding of multimodal and reinforcement-learning systems with a builder's instinct for building tools that hold up under real use, and is looking to bring that combination to production-grade agentic AI development.

Core Competencies

- **LLM Fine-Tuning & Evaluation:** Fine-tuned LLMs for applied QA tasks; basic DeepSpeed for distributed training; designed evaluation frameworks that surface model failure modes
- **Agentic AI Workflows:** Hands-on daily use of agentic coding assistants (Claude Code) for autonomous multi-step engineering tasks; LangChain, Autogen, FastAPI
- **Machine Learning & Deep Learning:** Python, PyTorch, TensorFlow, multimodal learning, graph neural networks, reinforcement learning (reward and policy design)
- **Research-to-Production:** Translating research outcomes into deployable prototypes and pipelines through academic-industry collaboration (Volkswagen)
- **Data Engineering:** Apache Spark, Hadoop (HDFS), Kafka, distributed RDF processing (SANSA Stack)

Professional Experience

DSIS, University of Bonn

Bonn, Germany

SENIOR RESEARCH SCIENTIST & PROJECT LEAD

2023-Present

- Designed and built an evaluation framework for large language models on spatial reasoning and geospatial question-answering, translating evaluation results into concrete model and pipeline improvements; fine-tuned LLMs for the task, with basic working experience using DeepSpeed for distributed training
- Leading the MoToRes AI research team, building a reinforcement-learning-based personalized route recommendation system from large-scale spatiotemporal data, including reward design and iterative experimentation
- Developed a multimodal deep learning architecture (PyTorch) fusing heterogeneous data sources into adaptive region-level embeddings, improving downstream prediction accuracy by 19%; published as first-author research at PAKDD 2024
- Built and maintained scalable ML pipelines integrating structured, unstructured, and knowledge-based data end-to-end, from ingestion through deployment-ready prototypes
- Collaborating with academic and industry partners to translate research outcomes into deployable analytical prototypes; supervising PhD, Master's, and Bachelor's students

L3S Research Center

Hannover, Germany

RESEARCH SCIENTIST

2019-2023

- Led the CampaNeo project, an industrial-academic AI collaboration with Volkswagen on large-scale predictive analytics for intelligent mobility
- Developed an RNN-based model for spatiotemporal accident prediction, improving performance by 4 percentage points through systematic benchmarking of ensemble and deep learning approaches
- Designed a privacy-preserving GPS trajectory watermarking framework (EU project SmashHit) achieving 99% recognition accuracy while preserving data utility

Fraunhofer IAIS

Sankt Augustin, Germany

DATA SCIENTIST

2018-2019

- Developed scalable distributed processing pipelines for large-scale geospatial and graph-based datasets (TomTom) using Spark and Scala
- Developed anomaly detection methods for knowledge graph analytics using the SANSA Stack and distributed RDF processing

Education

University of Bonn

Bonn, Germany

PHD IN COMPUTER SCIENCE / ARTIFICIAL INTELLIGENCE

2019-2025

- Thesis: "Adaptive Geospatial Data Representation for Data Analytics and Sharing in the Mobility Domain." Research spanning multimodal learning, representation learning, and graph neural networks. Magna cum laude (1.1)

- Thesis: "Scalable Numerical Outlier Detection in Knowledge Graphs," published at EKAW 2018. Grade: 1.9

- CGPA: 7.85/10

Selected Publications

- Rajjat Dadwal, Ran Yu, Elena Demidova (2024). A multimodal and multitask approach for adaptive geospatial region embeddings. PAKDD.
- Joshua Petteno, Rajjat Dadwal, Alishiba Dsouza, Elena Demidova (2026). MOSAIC: Synthetic Trajectory Generation with Reinforcement Learning. ACM SIGSPATIAL Poster Paper.
- Genivika Mann, Rajjat Dadwal, Elena Demidova (2026). DIRECT: Deep Reinforcement Learning for Tourist Route Generation. ACM TSAS Journal and SigSpatial conference (To appear).

Academic Service

2026-2027 **Program Committee Member**, ISWC, ECML PKDD, AAAI
2019-2025 **Reviewer**, NeurIPS, ICML, ISWC, ESWC, ACM SIGSPATIAL, WWW

Languages

- English: Fluent German: B1 Hindi: Native

Honors and Awards

- 2009 **Top 0.07% (Rank 78/100,000)**, National University Entrance Exam, National Institute of Technology, Hamirpur
- 2023 **Selected for a research visit**, University of Melbourne, Australia